Meet Aiara

A portal concept applying AI to promote civic engagement around the preservation and cleanup of rivers and other water bodies in the Brazil's Atlantic Forest.

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1. Meet Aiara

Research objective. Explore new possibilities for applying computational technologies to foster civic engagement and contribute to enriching the public and political debate regarding the health of rivers and other water bodies.

Methodology. Literature review of technology concepts and hydrological monitoring; semi-structured interviews with diverse experts; design thinking for scope definition and specifications; and development of a software prototype as a proof of concept.

Sample scope. 11 rivers from the Tietê River Basin were selected using public data from monitoring stations across various sources (Observando os Rios, HidroWeb, and CETESB).

What Aiara is. A portal conceived as an avatar or an **"artificial consciousness"** where users chat with an AI, while a "magic whiteboard" next to the chat displays the visual resource best suited to the question (map, chart, or metric). The name combines Iara, a **figure from Brazilian folklore** associated with the waters, with the acronyms AI and IA.

The concept. For strategic impact, we opted to develop a proof of concept (PoC) in the form of a controlled-access *beta* product tailored to the school community, involving students and educators. After all, they are the citizens of tomorrow, and a safe product for children and teenagers serves as both the means and foundation for a product aimed at the broader public. Its core features:

- **Focus on interpretation:** rather than replicating models of other environmental portals, such as MapBiomas, the goal is to answer questions like "how is the river near me doing?" and "has the river in my city improved over the last X years?".
- **Shared governance:** the maintaining organization defines the general boundaries of the AI, while the teacher adjusts the pedagogical usage within those limits.
- **Controlled interface.** The assistant does not generate unconstrained pages; instead, it indicates the block type to display (text, map, chart, metric), and the portal renders everything in a controlled manner. Conversations are logged anonymously in compliance with LGPD and the Digital ECA, and students are informed about which logs their teacher can access.

2. Conceptual interfaces

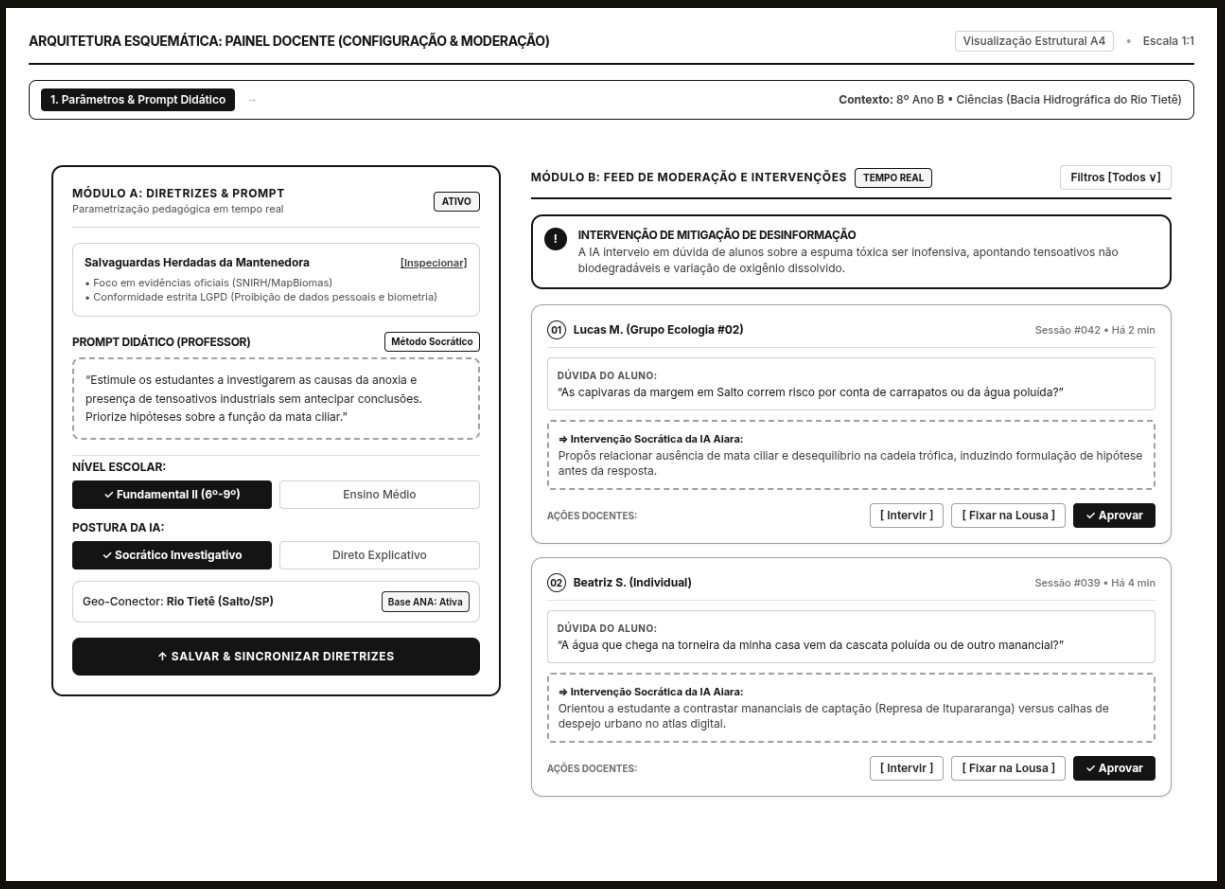
2.1 AI Parameterization by the Maintaining Organization



2.2 Student Interface (game-inspired)



2.3 Educator Moderation



3. How does it work?

Overview. Essentially, a data aggregation component gathers data from multiple sources (here we use HidroWeb/ANA, CETESB, and Observando os Rios) regarding different areas of the monitored rivers. The collected data is analyzed by an AI via API and pre-registered prompts, generating a structured summary report ("most relevant data") of current activity in each area of the river. Periodically, a background job iterates through the registered water resources (rivers, basins, or reservoirs), analyzes the structured summaries of the monitored points, and produces a "most relevant data" entry tied to each resource. Finally, the user frontend and backend (REST endpoints) of the portal leverage AI to analyze and interpret the various "most relevant data" entries and orchestrate the user interface.

Future potential. In its ideal formulation, Aiara would act as an "artificial consciousness" dedicated to river protection, consisting of AI agents capable of:

- multidimensionally analyzing data from various sources, including satellite imagery (riparian zone coverage), news, institutional reports, scientific research, etc.;
- engaging with citizens through a friendly and responsible persona;
- evolving under supervision, refining its responses based on feedback and new information;
- taking proactive actions (under human supervision), such as drafting emails, social media posts, and reports for distribution.